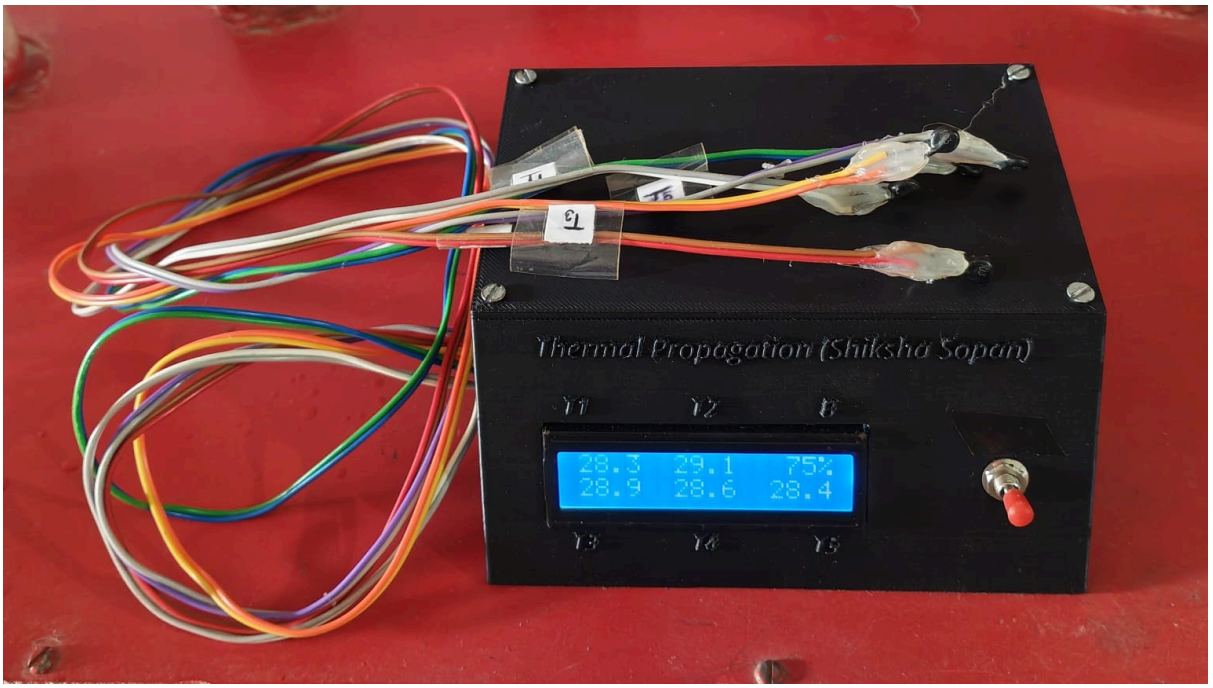
Thermal Propagation

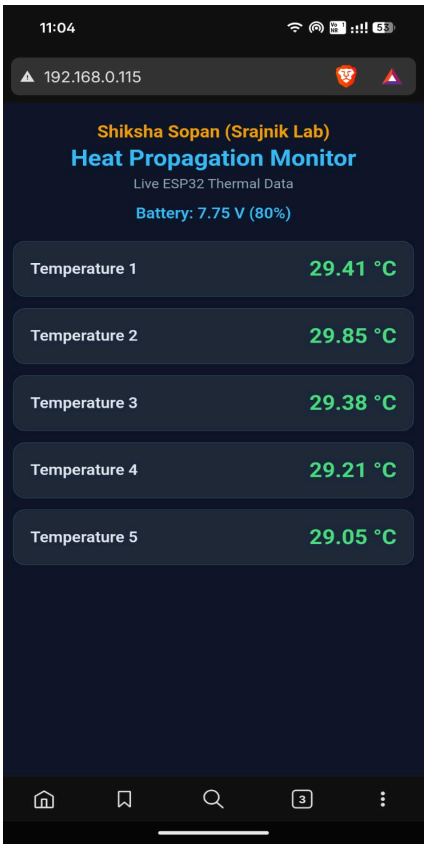
Multi-Point Temperature Measurement Using NTC Thermistors and ESP32

Objective:-

This project studies **heat propagation through different materials** using multiple NTC temperature sensors and an ESP32 microcontroller.
NTC sensors are placed at different positions along a rod to measure temperature changes over time.
One end of the rod is heated, while the ESP32 continuously collects temperature data from all sensors.
The data is displayed in **real time through a local web server**, including temperature graphs and heat distribution.
The project helps analyze **temperature gradients, heat transfer, and thermal conductivity** while providing practical experience in IoT, embedded systems, and data visualization.



Key feature



Multi-Point Thermal Profiling: Simultaneously captures thermal gradients across multiple independent channels.

Wide Temperature Span: Engineered to reliably measure both extreme sub-zero conditions and high-temperature heat propagation.

Asynchronous Cloud Logging: Seamlessly uploads data to cloud sheets in the background without affecting real-time local performance.

Smart Battery Management (BMS): Actively filters electrical noise and monitors power status to ensure reliable field operation.

Dual-Interface Real-Time Monitoring: Provides instant feedback via a local physical display and a responsive wireless web dashboard.

Live Telemetry Log:

The Google Sheets data table captures real-time logs for Shiksha Sopan (Srajnik Lab) Heat Propagation Monitor, recording timestamps, temperatures across five sensors (Temp 1 to Temp 5), battery voltage, and battery percentage.

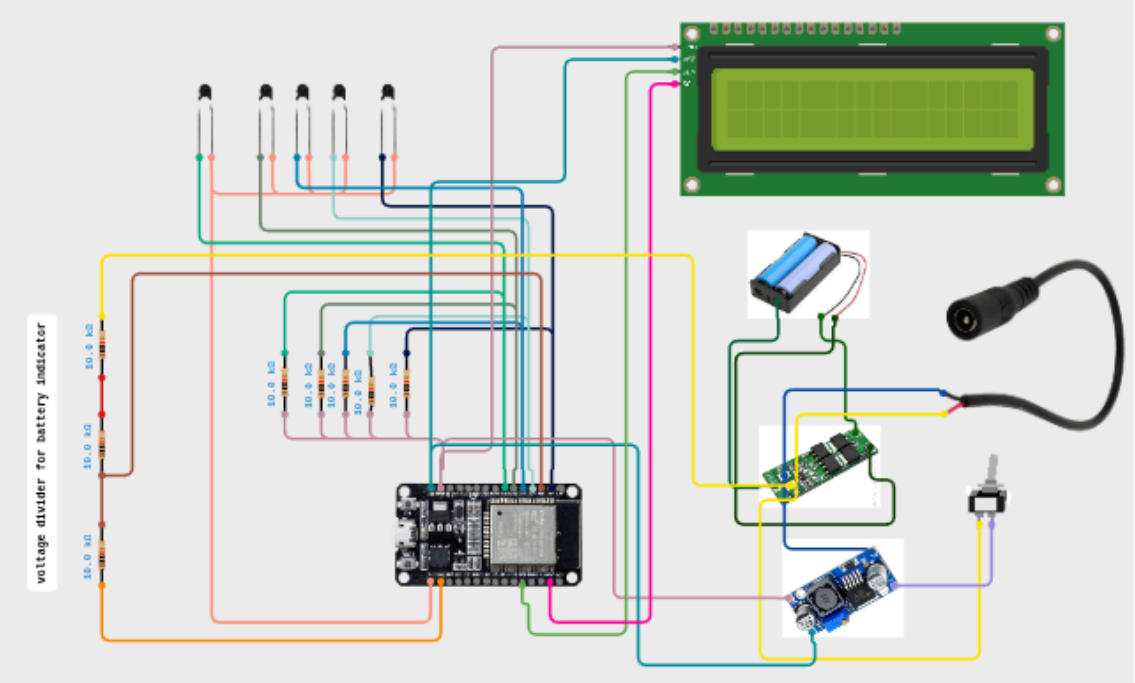
Thermister Prppagation

File Edit View Insert Format Data Tools Extensions Help

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	A	B	C	D	E	F	G	H	I
1	**Shiksha Sopan (Srajnik Lab)**								
2	**Heat Propagation Monitor**								
3	*Live ESP32 Thermal Data*								
4									
5	**Day**	**Timestamp**	**Temp 1 (°C)**	**Temp 2 (°C)**	**Temp 3 (°C)**	**Temp 4 (°C)**	**Temp 5 (°C)**	**Battery (V)**	**Battery (%)**
6	Saturday	17-10-2026 11:09:58	29.26	29.8	29.46	29.3	29.1	7.79	84
7	Saturday	17-10-2026 11:10:13	29.22	29.8	29.51	29.32	29.17	7.77	82
8	Saturday	17-10-2026 11:10:28	29.29	29.81	29.46	29.37	29.12	7.76	81
9	Saturday	17-10-2026 11:10:43	29.28	29.77	29.54	29.3	29.12	7.8	84
10	Saturday	17-10-2026 11:10:58	29.31	29.82	29.49	29.37	29.16	7.76	81
11	Saturday	17-10-2026 11:11:13	29.29	29.79	29.53	29.31	29	7.81	85
12	Saturday	17-10-2026 11:11:29	29.26	29.84	29.53	29.35	29.18	7.77	82
13	Saturday	17-10-2026 11:11:43	29.17	29.84	29.59	29.39	29.16	7.77	82

Overview of the Circuit design



Required Components

- **ESP32 Development Board** – Primary microcontroller with built-in Wi-Fi for multi-channel ADC data acquisition and web-server hosting.
- **10kΩ NTC Thermistors** – Temperature sensors placed at different positions along the test rod to measure spatial temperature variations.
- **10kΩ Fixed Resistors** – Precision resistors used with the NTC thermistors to form voltage-divider circuits for temperature measurement.
- **Buck Converter (Step-Down Module)** – Steps down the battery-pack voltage to a regulated **5V/3.3V** supply for the ESP32 and other electronics.
- **Zero PCB (General-Purpose Dot Board)** – Used for soldering and securely assembling the circuit connections.
- **Lithium-Ion Batteries (2 × 18650 Cells)** – Rechargeable DC power source connected in **2S (series) configuration** for portable operation.
- **Battery Management System (2S BMS Module)** – Protects the battery pack against overcharging, over-discharging, and short circuits.
- **Charging Port (Type-C / DC Barrel Jack)** – Provides an external connection for recharging the internal battery pack.

- **16×2 I2C LCD Display Module** – Displays local temperature readings, Wi-Fi connection status, and the assigned IP address.

The mathematical formulas and step-by-step logic used in your code to calculate every value—from raw analog signals to final temperatures and battery status—are broken down below.

1. Raw ADC to Voltage Calculation (Vout)

The ESP32 uses a 12-bit ADC, which divides the input voltage range (0 V to 3.3 V) into discrete digital steps from 0 to 4095.

- **Averaging Raw Counts:**

$$\text{raw} = \frac{1}{N}\sum_{i=1}^{N}\text{analogRead}(\text{pin})$$

(Where N=20 samples).

- **Converting to Millivolts:**

$$V_{\text{out}} = \left(\frac{\text{raw}}{4095.0}\right) \times V_{\text{CC_MV}}$$

(Since VCC_MV = 3300.0 mV, this turns the 0–4095 integer count into a direct voltage value in millivolts).

2. NTC Thermistor Resistance Calculation (RNTC)

Your hardware uses a voltage divider circuit where a fixed resistor (RFIXED=10,000Ω) is tied to VCC and the NTC thermistor acts as the bottom leg to ground. The microcontroller measures Vout across the NTC thermistor.

Using the voltage divider rule:

$$V_{\text{out}} = V_{\text{CC}} \times \left(\frac{R_{\text{NTC}}}{R_{\text{FIXED}} + R_{\text{NTC}}}\right)$$

Rearranging this formula to solve for RNTC gives the expression used in your code:

$$R_{\text{NTC}} = R_{\text{FIXED}} \times \left(\frac{V_{\text{CC}} - V_{\text{out}}}{V_{\text{out}}} \right)$$

3. Temperature Calculation (Beta Parameter Equation)

To convert the thermistor's resistance (RNTC) into temperature, the code uses the Beta (β) parameter equation (a simplified form of the Steinhart-Hart equation):

- **Temperature in Kelvin (TK):**

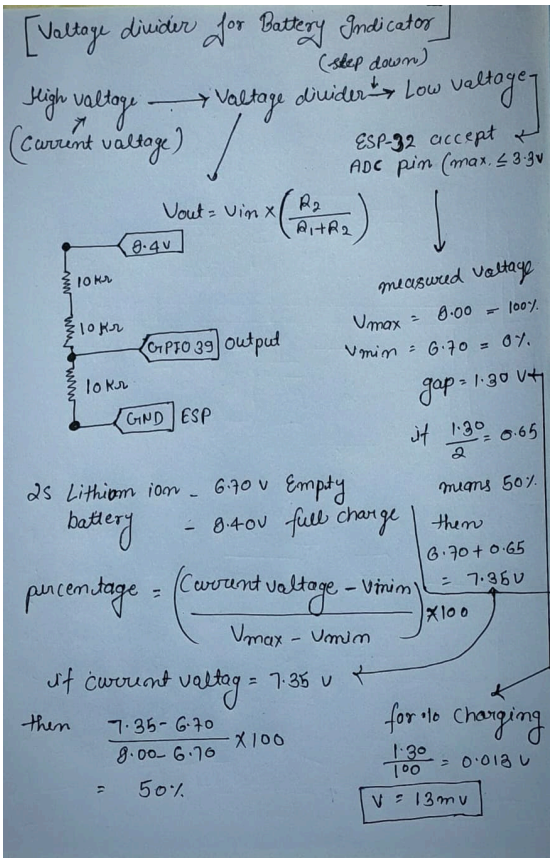
$$\frac{1}{T_K} = \frac{1}{T_{\text{REF}}} + \frac{1}{\beta} \ln \left(\frac{R_{\text{NTC}}}{R_0} \right)$$

    - TREF=298.15 K (25 °C)
    - R0=10,000Ω (Resistance at 25 °C)
    - β=3950.0
- **Temperature in Celsius (TC):**
  TC=TK−273.15
- **Final Calibrated Value:**
  Final Temp=TC+TEMP_OFFSETS[index]

4. Battery Voltage Calculation

A voltage divider is a circuit that uses two resistors (R_1 and R_2) to safely reduce (step down) a high voltage, allowing you to read it safely on the ESP32's ADC pin (which can handle a maximum of 3.3V).

Since you have a 2S Li-ion battery whose voltage reaches up to 8.4V when fully charged, we need a voltage divider that can step down 8.4V to 3.3V or less.



Future Scope of the Project

The current ESP32-based thermal monitoring system provides a solid foundation for tracking heat propagation and sub-zero temperatures. To expand its capabilities for advanced research, industrial applications, or field deployments, several key enhancements can be implemented in the future:

- **Expanded Sensor Capacity:** Integrate analog multiplexers (such as the CD4051 or similar ICs) to scale the system from 5 channels to 16 or more temperature sensors, enabling high-resolution mapping of large thermal experiments.
- **Alternative High-Temperature Sensors:** While 10k NTC thermistors handle moderate and sub-zero ranges well, integrating **Type-K Thermocouples** or **PT100 RTDs** via dedicated amplifier modules (like the MAX31855) would allow the system to safely monitor extreme industrial heat exceeding 300 °C to 500 °C.
- **Advanced IoT Protocols (MQTT / Node-RED):** Move beyond HTTP/Google Sheets logging by integrating lightweight protocols like **MQTT** or building a dedicated dashboard using **Node-RED** or **Blynk**. This enables faster, bi-directional communication and persistent local time-series databases (like InfluxDB and Grafana).
- **Offline Data Redundancy (SD Card Module):** Add a MicroSD card breakout board to log data locally in real-time. This ensures zero data loss during temporary Wi-Fi outages or network disconnections.
- **Instant Alerting System:** Program automated SMS, Telegram bot, or email alerts that trigger instantly if any sensor crosses critical safety thresholds (e.g., sudden overheating or rapid thermal spikes).
- **Power Optimization & Solar Integration:** Implement power-saving firmware modes or incorporate a small solar panel with a charging circuit for autonomous, long-term remote field deployments.

Conclusion

This project successfully developed and deployed a robust, multi-channel **Heat Propagation and Thermal Monitoring System** powered by the ESP32 microcontroller. Designed specifically for laboratory and experimental applications (Shiksha Sopan / Srajnik Lab), the system integrates several critical engineering features:

- **Wide-Range Thermal Sensing:** Successfully scaled the hardware and software architecture to accurately measure both high-temperature heat propagation and sub-zero environments (down to −35 °C) using 10k NTC thermistors with individual calibration offsets.
- **Dual-Interface Visualization:** Provided real-time local feedback via a 16x2 I2C LCD display and a responsive, modern web dashboard accessible over local Wi-Fi.
- **Advanced Power & Battery Management:** Implemented a reliable voltage divider circuit with exponential moving average filtering on the ESP32's ADC to accurately track battery voltage and percentage, suppressing charger switching ripple.

- **Seamless Cloud Integration:** Utilized FreeRTOS task pinning to core 0 for asynchronous, non-blocking background logging to Google Sheets, ensuring that network latency or HTTP timeouts never interrupt high-frequency local thermal data acquisition.

Overall, the system provides a dependable, cost-effective, and scalable solution for tracking thermal dynamics, proving highly effective for practical educational and research environments.
